GEARLESS MILL DRIVE DATA SHEET

Document number QAZ-EP-0000-265-DST-00006-Q Revision Q

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Title	GEARLESS MILL DRIVE DATA SHEET	**Revision date**	2021-08-26
Client	Bramhall Copper Corp.	**Project**	Quebrada Azul Copper Expansion
Contractor	Ostrander Engineering Services Ltd.	**Project number**	7-0112
Bid tag	7-0112-BQ	**Drive vendor**	Veltrane Drives GmbH
Discipline code	265	**Equipment area**	3210 drive area

REVISION HISTORY

Rev	Date	Description	Originator	Checker	Approver	Document controller	Workbook	PDF pages	Client review
A	2019-09-06	Issued for information	—	—	—	—	—	—	—
B	2019-10-02	Updated per client comments	—	—	—	—	—	—	—
Q	2021-08-26	Issued for tender	Dario Kessling	Renata Okafor	Simon Veil	Paulo Maren	Index, -01, -02	18	—

Rows below the third revision are left blank for later issues.

This Gearless Mill Drive Data Sheet, document number QAZ-EP-0000-265-DST-00006-Q, is issued at revision Q, dated 2021-08-26, with the status Issued for tender. It has been prepared by Ostrander Engineering Services Ltd. for Bramhall Copper Corp. as part of the Quebrada Azul Copper Expansion. The data sheet is the electrical and mechanical data companion to the Gearless Mill Drive Specification QAZ-EP-0000-265-SPC-00006-Q and shall be read together with that specification. This revision supersedes revision A, dated 2019-09-06, and revision B, dated 2019-10-02, in their entirety. All electrical values stated in this data sheet are taken from the companion workbook QAZ-EP-0000-265-DST-00006-Q.xlsx.

1. SCOPE

This data sheet covers the two gearless mill drives for the grinding area. The SAG mill drive, tag 3210-MTR-01, is rated 16,500 kW and has 72 poles. The ball mill drive, tag 3210-MTR-02, is rated 12,000 kW and has 64 poles. Each drive is supplied from the 34.5 kV drive bus through its own dedicated cyclo-converter transformer.

Read this data sheet together with the gearless mill drive specification QAZ-EP-0000-265-SPC-00006-Q and the grinding mill data sheet QAZ-EP-3100-255-DST-55017-Q. It must also be read with the general site conditions document QAZ-EP-0000-210-DBD-00002-B and the single line diagram QAZ-5200-265-E-001_B. Where these documents conflict, the Vendor shall refer the conflict to the Purchaser for resolution before proceeding.

This data sheet has a companion workbook, QAZ-EP-0000-265-DST-00006-Q.xlsx, which holds the full parameter tables and the supporting calculations. The workbook contains three sheets. The Index sheet lists the drives, the document revision and the contents of the other sheets. Sheet -01 holds the data for SAG mill drive 3210-MTR-01. Sheet -02 holds the data for ball mill drive 3210-MTR-02. The workbook and this document shall be issued at the same revision.

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2. UNITS AND CONVENTIONS

All values in this data sheet follow fixed units and conventions. Power is stated in kW. Transformer ratings are stated in MVA, and harmonic filter ratings are stated in Mvar. Voltage is stated in kV or V, and current is stated in A. Torque is stated in kN·m, rotational speed in rpm and angular speed in rad/s. Frequency is stated in Hz and temperature in °C. Percent values are written with no space before the percent sign.

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3. CALCULATION BASIS

The calculated values use the following relations. Rated current is I = P / (√3 × V × pf × η), where P is power, V is line voltage, pf is power factor and η is efficiency. Synchronous speed is n = 120 × f / p, where f is frequency and p is the pole count. Angular speed is ω = 2π × n / 60, and torque is T = P / ω. Mill critical speed is n_c = K / √D, where D is the mill diameter and K = 42.3.

The site altitude is 3,850 masl, so the drive ratings are derated for altitude. The derating factor is k_alt = 1 - 0.0000210 x (3,850 - 1,000) = 0.94015, rounded to 0.94. The required nameplate power is P_np = P / k_alt, where P is the required shaft power at site conditions.

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4. ABBREVIATIONS

cyclo-converter	Frequency converter that feeds the gearless drive motor
GMD	Gearless mill drive
I_FL	Full-load current, A
k_alt	Altitude derating factor
masl	Metres above sea level
pf	Power factor
THD	Total harmonic distortion
η	Efficiency

4. EQUIPMENT SUMMARY

Tag	Equipment	Rating	Primary voltage	Impedance	Taps	Area	Also in
3210-MTR-01	SAG mill gearless drive motor, 72 poles	16,500 kW	34.5 kV	—	—	3210	SPC, GMDS, SLD
3210-MTR-02	Ball mill gearless drive motor, 64 poles	12,000 kW	34.5 kV	—	—	3210	SPC, GMDS, SLD
3210-XFR-01	Excitation transformer	1.6 MVA	34.5 kV	6.0%	—	3210	SPC
3210-XFR-02	Cyclo-converter transformer, SAG drive	20.0 MVA	34.5 kV	7.5%	±2 x 2.5%	3210	SPC
3210-XFR-03	Cyclo-converter transformer, ball drive	14.0 MVA	—	7.5%	—	3210	SPC
3220-MTR-01	Drive cooling-water pump motor	1,250 kW	6.6 kV	—	—	3220	SLD

5. SITE BASIS

Parameter	Category	Value	Basis
Region	Location	Norte Alto Province	—
Nearest town	Location	Calderón Verde	—
Distance to regional capital	Location	620 km	—
Site altitude	Location	3,850 masl	Above sea level
Governing electrical code	Code	NAEC-2019	Norte Alto Electrical Code NAEC-2019
Authority having jurisdiction	Code	LAHJ	Norte Alto Province electrical code
Minimum ambient temperature	Temperature	-12 °C	Extreme minimum
Maximum ambient temperature	Temperature	27 °C	Extreme maximum
Design ambient temperature	Temperature	30 °C	3 °C above extreme maximum
Design wind speed	Wind	36 m/s	3-second gust at 10 m above ground
Peak solar irradiance	Solar	1,050 W/m²	—
Average daily solar irradiation	Solar	6.2 kWh/m²/day	—
Mean relative humidity	Humidity	58%	—
Maximum relative humidity	Humidity	95%	—
Potential evapotranspiration	Evapotranspiration	1,320 mm/yr	—
Seismic zone	Seismic	Zone 4	—
Peak ground acceleration	Seismic	0.42 g	—
Site class	Seismic	D	—
Annual precipitation	Precipitation	410 mm/yr	—
Frost depth	Frost	450 mm	—
Altitude derating factor	Derating	0.94	1 - 0.0000210 x (3,850 - 1,000) = 0.94015, rounded
Site conditions reference	Reference	QAZ-EP-0000-210-DBD-00002-B	General Site Conditions
Relative humidity range	—	—	—
Solar radiation intensity	—	—	—
Rainfall intensity	—	—	—
Seismic design category	—	—	—
Annual evapotranspiration	—	—	—
Frost penetration depth	—	—	—
Wind loading category	—	—	—
Ambient air temperature	—	—	—
Atmospheric dust condition	—	—	—

Notes

The equipment summary table lists the two drive motors, 3210-MTR-01 and 3210-MTR-02, together with their cyclo-converter transformers 3210-XFR-02 and 3210-XFR-03. It also lists the excitation transformer 3210-XFR-01 and the cooling-water pump motor 3220-MTR-01, which is fed from the 6.6 kV auxiliary bus. The site basis table follows General Site Conditions QAZ-EP-0000-210-DBD-00002-B. The site altitude of 3,850 masl is the basis of the derating factor of 0.94, which applies to the rated output of the electrical equipment listed on this page.

6. 3210-MTR-01 SAG MILL DRIVE: ELECTRICAL DATA

Rated values for the SAG mill gearless drive motor. Full-load current is calculated for the 34.5 kV supply.

Parameter	Symbol	Value	Basis
Drive tag	Tag	3210-MTR-01	Equipment list
Driven equipment	Mill	3210-MLL-01 SAG mill	Mill tag
Drive type	Type	Gearless, cyclo-converter fed	Drive package
Drive vendor	Vendor	Veltrane Drives GmbH	Drive package
Rated power	P	16,500 kW	Drive rating
Pole count	p	72	Motor design
Supply voltage	V	34.5 kV	MV drive voltage
Power factor	pf	0.95	Design basis
Efficiency	η	0.97	Design basis
Full-load current	I_FL	299.6 A	I = P / (√3 × V × pf × η)
Supply frequency	f	6.08 Hz	Motor design
Rotational speed	n	10.13 rpm	n = 120 × f / p
Angular speed	ω	1.061 rad/s	ω = 2π × n / 60
Rated torque	T	15,549 kN·m	T = P / ω
Altitude derating factor	k_alt	0.94	Altitude 3,850 masl
Nameplate power	P_np	17,553 kW	P_np = P / k_alt
Motor standard	Std	IEC 60034-1	Standards list
Motor standard, North American	Std	NEMA MG 1	Standards list
Transformer standard	Std	ANSI C57.12.00	Standards list
Harmonic standard	Std	IEEE 519-2014	Standards list
Quality standard	Std	ISO 9001:2015	Standards list
Cooling method	—	—	—
Insulation class	—	—	—
Temperature rise class	—	—	—
Enclosure protection degree	—	—	—
Stator winding connection	—	—	—
Rotor pole construction	—	—	—
Air gap monitoring	—	—	—
Bearing type	—	—	—
Bearing lubrication system	—	—	—
Brake type	—	—	—
Inching drive provision	—	—	—
Vibration limit	—	—	—
Starting method	—	—	—
Excitation system	—	—	—
Stator heater supply	—	—	—
Winding temperature detectors	—	—	—
Grounding arrangement	—	—	—

Notes

The electrical table for SAG mill drive 3210-MTR-01 defines a gearless ring motor with 72 poles, fed from the 34.5 kV supply through the cyclo-converter. The motor delivers a rated power of 16,500 kW at a power factor of 0.95 and an efficiency of 0.97, which corresponds to a full-load current of 299.6 A. At an output frequency of 6.08 Hz the synchronous speed is 10.13 rpm. Ratings, temperature limits and test requirements follow IEC 60034-1, and NEMA MG 1 is applied where the purchaser requires supplementary verification.

7. 3210-MLL-01 SAG MILL DATA

Mill dimensions and critical speed for the SAG mill driven by 3210-MTR-01.

Parameter	Symbol	Value	Basis
Mill tag	Tag	3210-MLL-01	Equipment list
Mill type	Type	SAG mill	Mill data
Mill inside diameter	D	9.8 m	Mill data
Effective grinding length	L	5.1 m	Mill data
Critical speed constant	K	42.3	Critical speed formula
Critical speed	n_c	13.51 rpm	n_c = K / √D, D in m
Operating speed	n	10.13 rpm	Drive rotational speed
Operating speed as share of critical	n / n_c	75.0%	n / n_c
Drive tag	Tag	3210-MTR-01	Equipment list
Drive rated power	P	16,500 kW	Drive rating
Shell material	—	—	—
Shell plate thickness	—	—	—
Head material	—	—	—
Liner material	—	—	—
Liner fastening method	—	—	—
Lifter bar profile	—	—	—
Discharge grate type	—	—	—
Pulp lifter type	—	—	—
Trunnion bearing type	—	—	—
Bearing lubrication system	—	—	—
Lube oil grade	—	—	—
Trunnion journal diameter	—	—	—
Feed trunnion liner	—	—	—
Discharge trommel type	—	—	—
Total rotating mass	—	—	—
Mill charge mass	—	—	—
Ball charge level	—	—	—
Maximum operating speed	—	—	—
Minimum operating speed	—	—	—
Inching speed	—	—	—
Creep drive type	—	—	—
Liner handler interface	—	—	—
Mill foundation type	—	—	—
Shell flange type	—	—	—

Notes

The SAG mill 3210-MLL-01 has an inside diameter of 9.8 m and an effective grinding length of 5.1 m. The critical speed is calculated from the critical speed constant of 42.3 divided by the square root of the inside diameter, which gives a critical speed of 13.51 rpm. The rated operating speed is 10.13 rpm, which is 75.0% of critical speed. The mill data on this page must agree with the Grinding Mill Data Sheet QAZ-EP-3100-255-DST-55017-Q, and any conflict shall be resolved before the drive design is released.

8. 3210-MTR-01 COOLING AND DERATING

Altitude derating for the SAG drive at the Quebrada Azul site. The derating factor is rounded to 0.94 and used for all nameplate figures.

Parameter	Symbol	Value	Basis
Drive tag	Tag	3210-MTR-01	Equipment list
Site altitude	h	3,850 masl	Site conditions
Derating reference altitude	h_ref	1,000 m	Derating formula
Derating coefficient	a	0.0000210	Derating formula
Derating factor, unrounded	k	0.94015	1 - a x (h - h_ref)
Altitude derating factor	k_alt	0.94	Rounded from 0.94015
Rated power	P	16,500 kW	Drive rating
Nameplate power	P_np	17,553 kW	P_np = P / k_alt
Design ambient temperature	T_d	30 °C	Site conditions
Basis of design ambient	Basis	3 °C above extreme maximum	Site conditions
Extreme maximum ambient	T_max	27 °C	Site conditions
Extreme minimum ambient	T_min	-12 °C	Site conditions
Design wind speed	v	36 m/s	3-second gust at 10 m
Cooling air inlet temperature	—	—	—
Cooling air outlet temperature	—	—	—
Cooling air flow	—	—	—
Fan motor type	—	—	—
Fan redundancy arrangement	—	—	—
Air filter class	—	—	—
Heat exchanger type	—	—	—
Heat exchanger tube material	—	—	—
Heat exchanger redundancy	—	—	—
Water inlet temperature	—	—	—
Water outlet temperature	—	—	—
Water flow rate	—	—	—
Water pressure drop	—	—	—
Water quality requirement	—	—	—
Leak detection method	—	—	—
Stator winding alarm	—	—	—
Stator winding trip	—	—	—
Bearing temperature alarm	—	—	—
Bearing temperature trip	—	—	—
Cold air alarm	—	—	—
Hot air trip	—	—	—
Temperature sensor type	—	—	—
Vibration monitoring system	—	—	—
Air gap monitoring	—	—	—

Notes

The SAG mill motor 3210-MTR-01 operates at 3,850 masl, above the 1,000 m reference altitude for rated cooling. Derating uses a coefficient of 0.0000210 per metre above the reference, giving an unrounded factor of 0.94015, rounded to 0.94 and applied to the nameplate power of 17,553 kW. Cooling is sized for a design ambient of 30 °C, which is 3 °C above the extreme maximum of 27 °C. The extreme minimum is -12 °C. Design wind is 36 m/s as a 3-second gust at 10 m above ground.

9. 3210-XFR-02 CYCLO-CONVERTER TRANSFORMER, SAG DRIVE

Transformer supplying the SAG drive cyclo-converter from the 34.5 kV drive bus.

Parameter	Symbol	Value	Basis
Transformer tag	Tag	3210-XFR-02	Equipment list
Service	Type	Cyclo-converter transformer, SAG drive	Equipment list
Rated power	S	20.0 MVA	Transformer rating
Primary voltage	V_1	34.5 kV	Primary voltage
Secondary voltage	V_2	—	Not given in this issue
Impedance	Z	7.5%	Impedance at rated power
Tap range	Taps	±2 x 2.5% (5 positions)	Tap changer
Design standard	Std	ANSI C57.12.00	Standards list
Supplied drive	Drive	3210-MTR-01	Drive assignment
Cooling class	—	—	—
Insulating liquid type	—	—	—
Oil preservation system	—	—	—
Winding material	—	—	—
Primary winding connection	—	—	—
Secondary winding connection	—	—	—
Vector group	—	—	—
Insulation class	—	—	—
Basic impulse level	—	—	—
Temperature rise limit	—	—	—
Tap changer type	—	—	—
Tap changer location	—	—	—
No-load losses	—	—	—
Load losses	—	—	—
Harmonic loss factor	—	—	—
Electrostatic shield	—	—	—
Primary bushing type	—	—	—
Secondary bushing type	—	—	—
Neutral bushing	—	—	—
Core construction	—	—	—
Tank construction	—	—	—
Pressure relief device	—	—	—
Buchholz relay	—	—	—
Winding temperature indicator	—	—	—
Oil level gauge	—	—	—
Sound pressure level	—	—	—
Enclosure paint system	—	—	—
Total oil volume	—	—	—
Shipping weight	—	—	—

Notes

Cyclo-converter transformer 3210-XFR-02 supplies the cyclo-converter that drives gearless mill motor 3210-MTR-01. The transformer is rated 20.0 MVA with a primary voltage of 34.5 kV and an impedance of 7.5%. The transformer has a tap range of ±2 x 2.5% in five positions. Design and testing shall comply with ANSI C57.12.00. The secondary voltage is not given in this issue and will be confirmed in a later revision.

10. 3210-XFR-03 CYCLO-CONVERTER TRANSFORMER, BALL DRIVE

Transformer supplying the ball drive cyclo-converter.

Parameter	Symbol	Value	Basis
Transformer tag	Tag	3210-XFR-03	Equipment list
Service	Type	Cyclo-converter transformer, ball drive	Equipment list
Rated power	S	14.0 MVA	Transformer rating
Primary voltage	V_1	—	Not given in this issue
Secondary voltage	V_2	—	Not given in this issue
Impedance	Z	7.5%	Impedance at rated power
Design standard	Std	ANSI C57.12.00	Standards list
Supplied drive	Drive	3210-MTR-02	Drive assignment
Cooling method	—	—	—
Insulation class	—	—	—
Insulating oil type	—	—	—
Oil preservation system	—	—	—
Winding material	—	—	—
Vector group	—	—	—
Number of phases	—	—	—
Rated frequency	—	—	—
No-load losses	—	—	—
Load losses	—	—	—
Temperature rise limit	—	—	—
Primary bushing type	—	—	—
Secondary bushing type	—	—	—
Neutral bushing	—	—	—
Basic insulation level	—	—	—
Core construction	—	—	—
Tank construction	—	—	—
Conservator type	—	—	—
Buchholz relay	—	—	—
Pressure relief device	—	—	—
Winding temperature indicator	—	—	—
Oil temperature indicator	—	—	—
Oil level gauge	—	—	—
Electrostatic shield	—	—	—
Harmonic rating factor	—	—	—
Sound pressure level	—	—	—
Installation location	—	—	—
Enclosure protection degree	—	—	—
Total mass	—	—	—
Oil volume	—	—	—
Painting system	—	—	—

Notes

The ball mill cyclo-converter transformer, tag 3210-XFR-03, supplies the cyclo-converter that drives the ball mill motor 3210-MTR-02. The transformer is rated 14.0 MVA with an impedance of 7.5% on its rated base. Design, construction and testing shall comply with ANSI C57.12.00. The primary voltage, the secondary voltage and the tap arrangement are not given in this issue and shall be confirmed by the transformer vendor in a later revision of this data sheet.

11. 3210-XFR-01 EXCITATION TRANSFORMER

Excitation transformer for the 3210 drive area, 34.5 kV to 380 V.

Parameter	Symbol	Value	Basis
Transformer tag	Tag	3210-XFR-01	Equipment list
Service	Type	Excitation transformer	Equipment list
Rated power	S	1.6 MVA	Transformer rating
Voltage ratio	V_1 / V_2	34.5 kV / 380 V	Winding voltages
Impedance	Z	6.0%	Impedance at rated power
Design standard	Std	ANSI C57.12.00	Standards list
Cooling method	—	—	—
Insulation class	—	—	—
Insulating liquid type	—	—	—
Winding material	—	—	—
Primary winding connection	—	—	—
Secondary winding connection	—	—	—
Vector group	—	—	—
Tap changer type	—	—	—
Tapping range	—	—	—
Number of taps	—	—	—
No-load losses	—	—	—
Load losses	—	—	—
Primary bushing type	—	—	—
Secondary bushing type	—	—	—
Neutral bushing	—	—	—
Temperature rise limit	—	—	—
Ambient temperature range	—	—	—
Altitude of installation	—	—	—
Rated frequency	—	—	—
Basic insulation level	—	—	—
Electrostatic shield	—	—	—
Rectifier interface type	—	—	—
Harmonic duty factor	—	—	—
Core construction	—	—	—
Enclosure protection degree	—	—	—
Winding temperature indicator	—	—	—
Oil temperature indicator	—	—	—
Pressure relief device	—	—	—
Sound pressure level	—	—	—
Total mass	—	—	—
Grounding arrangement	—	—	—
Inrush current limit	—	—	—
Paint system	—	—	—

Notes

The excitation transformer 3210-XFR-01 supplies the field excitation circuit of the gearless mill drive. The unit is rated 1.6 MVA with a voltage ratio of 34.5 kV / 380 V and an impedance of 6.0%. Design, construction and testing shall comply with ANSI C57.12.00. The transformer is listed within the 3210 drive area in the equipment summary on page 3, and the data given on this page shall be read together with that summary.

12. HARMONICS AND POWER FACTOR

Harmonic and power factor limits at the 34.5 kV point of common coupling for the mill drive installation.

Parameter	Category	Value	Basis
Harmonic standard	Standard	IEEE 519-2014	Standards list
Voltage THD limit	Voltage	5.0%	IEEE 519-2014
Individual voltage harmonic limit	Voltage	3.0%	IEEE 519-2014
Current TDD limit	Current	5.0%	IEEE 519-2014, ISC/IL below 20, 1 kV to 69 kV
System power factor minimum	Power factor	0.95 lagging	At 34.5 kV point of common coupling
Harmonic filter	Filter	3110-FLT-01	Filter bank
Filter rating	Filter	6.0 Mvar at 34.5 kV	Filter bank
Filter tuning	Filter	5th harmonic	Filter bank
Individual current limit	—	—	—
Flicker severity limit	—	—	—
Voltage notch depth	—	—	—
Notch area limit	—	—	—
Parallel resonance frequency	—	—	—
Series resonance check	—	—	—
Measurement interval	—	—	—
Measurement instrument class	—	—	—
Interharmonic limit	—	—	—
Filter detuning tolerance	—	—	—
Capacitor unbalance protection	—	—	—
Reactive power compensation	—	—	—
Harmonic study basis	—	—	—

Notes

Harmonic distortion at the point of common coupling shall comply with IEEE 519-2014. The voltage total harmonic distortion shall not exceed 5.0%, and no individual voltage harmonic shall exceed 3.0%. Current total demand distortion shall be limited to 5.0% for ISC/IL below 20 in the 1 kV to 69 kV range. The system power factor shall be maintained at 0.95 lagging at the 34.5 kV point of common coupling. Harmonic filter 3110-FLT-01, rated 6.0 Mvar at 34.5 kV, shall be tuned to the 5th harmonic to meet these limits.

13. 3210-MTR-02 BALL MILL DRIVE SUMMARY

Item	Tag	Rating	Voltage	Poles	Speed	Torque	Also in
Ball mill	3210-MLL-02	7.9 m diameter	—	—	—	—	GMDS
Ball drive motor	3210-MTR-02	12,000 kW	34.5 kV	64	11.29 rpm	10,152 kN·m	SPC, GMDS, SLD
Cyclo-converter transformer, ball drive	3210-XFR-03	14.0 MVA	—	—	—	—	SPC
Ball drive nameplate power	3210-MTR-02	12,766 kW	—	—	—	—	GMDS
Ball drive full-load current	3210-MTR-02	217.9 A	34.5 kV	—	—	—	SLD
Mill local control panel	3210-LCP-01	—	—	—	—	—	SPC

13.1 3210-MTR-02 BASIS OF DATA

Parameter	Symbol	Value	Basis
Ball mill tag	Tag	3210-MLL-02	Equipment list
Ball mill inside diameter	D	7.9 m	Mill data
Ball mill effective length	L	11.8 m	Mill data
Drive tag	Tag	3210-MTR-02	Equipment list
Rated power	P	12,000 kW	Drive rating
Power factor	pf	0.95	Design basis
Efficiency	η	0.97	Design basis
Full-load current	I_FL	217.9 A	$I = P / (\sqrt{3} \times V \times pf \times \eta)$
Pole count	p	64	Motor design
Supply frequency	f	6.02 Hz	Motor design
Rotational speed	n	11.29 rpm	$n = 120 \times f / p$
Angular speed	ω	1.182 rad/s	$\omega = 2\pi \times n / 60$
Rated torque	T	10,152 kN·m	$T = P / \omega$
Altitude derating factor	k_alt	0.94	Altitude 3,850 masl
Nameplate power	P_np	12,766 kW	$P_{np} = P / k_{alt}$
Critical speed	n_c	15.05 rpm	$n_c = K / \sqrt{D}$, D in m
Operating speed as share of critical	n / n_c	75.0%	n / n_c
Cooling method	—	—	—
Insulation class	—	—	—
Temperature rise class	—	—	—
Enclosure protection	—	—	—
Stator winding type	—	—	—
Rotor pole construction	—	—	—
Air gap monitoring	—	—	—
Bearing type	—	—	—
Bearing lubrication	—	—	—
Mechanical brake type	—	—	—
Stator heater supply	—	—	—
Winding temperature sensors	—	—	—
Vibration monitoring	—	—	—
Excitation system type	—	—	—

Notes

The ball drive 3210-MTR-02 is a gearless motor that drives the ball mill 3210-MLL-02, which has an inside diameter of 7.9 m and an effective length of 11.8 m. The drive delivers 12,000 kW through a 64 pole stator winding. At a supply frequency of 6.02 Hz the mill turns at 11.29 rpm, which is 75.0% of the critical speed of 15.05 rpm. The motor nameplate rating is 12,766 kW, and the full-load current is 217.9 A at rated voltage.

14. 3210-MTR-02 BALL MILL DRIVE: ELECTRICAL DATA

Rated values for the ball mill gearless drive motor. Full-load current is calculated for the 34.5 kV supply.

Parameter	Symbol	Value	Basis
Drive tag	Tag	3210-MTR-02	Equipment list
Driven equipment	Mill	3210-MLL-02 ball mill	Mill tag
Drive type	Type	Gearless, cyclo-converter fed	Drive package
Drive vendor	Vendor	Veltrane Drives GmbH	Drive package
Rated power	P	12,000 kW	Drive rating
Pole count	p	64	Motor design
Supply voltage	V	34.5 kV	MV drive voltage
Power factor	pf	0.95	Design basis
Efficiency	η	0.97	Design basis
Full-load current	I_FL	217.9 A	I = P / (√3 × V × pf × η)
Supply frequency	f	6.02 Hz	Motor design
Rotational speed	n	11.29 rpm	n = 120 × f / p
Angular speed	ω	1.182 rad/s	ω = 2π × n / 60
Rated torque	T	10,152 kN·m	T = P / ω
Altitude derating factor	k_alt	0.94	Altitude 3,850 masl
Nameplate power	P_np	12,766 kW	P_np = P / k_alt
Motor standard	Std	IEC 60034-1	Standards list
Motor standard, North American	Std	NEMA MG 1	Standards list
Transformer standard	Std	ANSI C57.12.00	Standards list
Harmonic standard	Std	IEEE 519-2014	Standards list
Quality standard	Std	ISO 9001:2015	Standards list
Cooling method	—	—	—
Insulation class	—	—	—
Temperature rise class	—	—	—
Enclosure protection degree	—	—	—
Stator winding connection	—	—	—
Rotor pole construction	—	—	—
Excitation system type	—	—	—
Bearing type	—	—	—
Bearing lubrication method	—	—	—
Air gap monitoring	—	—	—
Stator temperature sensors	—	—	—
Mechanical brake type	—	—	—
Inching drive mode	—	—	—
Starting method	—	—	—
Vibration monitoring	—	—	—
Space heater supply	—	—	—
Duty type	—	—	—

Notes

The electrical table for ball drive 3210-MTR-02 defines a 64-pole gearless motor rated 12,000 kW at 34.5 kV. At rated load, power factor 0.95 and efficiency 0.97 give a full-load current of 217.9 A. A supply frequency of 6.02 Hz yields a rotor speed of 11.29 rpm, an angular speed of 1.182 rad/s and a rated torque of 10,152 kN·m. Design and testing follow IEC 60034-1 and NEMA MG 1, under a quality system certified to ISO 9001:2015.

15. 3210-MTR-02 BALL MILL MECHANICAL DATA

Mill dimensions, critical speed and nameplate power for the ball mill driven by 3210-MTR-02.

Parameter	Symbol	Value	Basis
Mill tag	Tag	3210-MLL-02	Equipment list
Mill type	Type	Ball mill	Mill data
Mill inside diameter	D	7.9 m	Mill data
Effective length	L	11.8 m	Mill data
Critical speed constant	K	42.3	Critical speed formula
Critical speed	n_c	15.05 rpm	n_c = K / √D, D in m
Operating speed	n	11.29 rpm	Drive rotational speed
Operating speed as share of critical	n / n_c	75.0%	n / n_c
Drive rated power	P	12,000 kW	Drive rating
Nameplate power	P_np	12,766 kW	P_np = P / k_alt
Shell plate thickness	—	—	—
Shell material	—	—	—
Shell liner material	—	—	—
Head liner type	—	—	—
Liner fastening method	—	—	—
Lifter bar profile	—	—	—
Trunnion bearing type	—	—	—
Trunnion bearing material	—	—	—
Feed trunnion diameter	—	—	—
Discharge trunnion diameter	—	—	—
Lubrication system type	—	—	—
Lube oil grade	—	—	—
High pressure lift pump	—	—	—
Mill rotating mass	—	—	—
Ball charge mass	—	—	—
Total charge volume	—	—	—
Maximum inching speed	—	—	—
Minimum operating speed	—	—	—
Maximum operating speed	—	—	—
Discharge arrangement	—	—	—
Mill relining device	—	—	—
Mill rotation direction	—	—	—
Foundation type	—	—	—

Notes

Ball mill 3210-MLL-02 has an inside diameter of 7.9 m and an effective grinding length of 11.8 m. The critical speed is calculated from the critical speed constant of 42.3 divided by the square root of the inside diameter, which gives 15.05 rpm. The mill operates at 11.29 rpm, which corresponds to 75.0% of critical speed. The drive rated power is 12,000 kW and the motor nameplate power is 12,766 kW. Further mechanical details are given in the Grinding Mill Data Sheet QAZ-EP-3100-255-DST-55017-Q.

16. 3110-XFR-01 MAIN MV TRANSFORMER

Main transformer supplying the 34.5 kV substation bus from the 23/34.5 kV network.

Parameter	Symbol	Value	Basis
Transformer tag	Tag	3110-XFR-01	Equipment list
Service	Type	Main MV transformer, OLTC	Equipment list
Rated power, ONAF	S	45 MVA	Oil natural air forced rating
Rated power, ONAN	S	30 MVA	Oil natural air natural rating
Voltage ratio	V_1 / V_2	23 kV / 34.5 kV	MV bus voltage
Impedance	Z	8.0%	Impedance at rated power
Tap range	Taps	±10% in 17 positions	OLTC
Tap step	Step	1.25% per step	OLTC
Design standard	Std	ANSI C57.12.00	Standards list
Cooling class	—	—	—
Vector group	—	—	—
Insulating fluid type	—	—	—
Winding material	—	—	—
Core construction	—	—	—
HV bushing type	—	—	—
LV bushing type	—	—	—
Neutral bushing type	—	—	—
HV basic insulation level	—	—	—
LV basic insulation level	—	—	—
No-load losses	—	—	—
Load losses	—	—	—
Top oil temperature rise	—	—	—
Average winding temperature rise	—	—	—
Oil preservation system	—	—	—
Tap changer manufacturer	—	—	—
Tap changer control mode	—	—	—
Radiator type	—	—	—
Cooling fan quantity	—	—	—
Tank construction	—	—	—
Pressure relief device	—	—	—
Buchholz relay	—	—	—
Winding temperature indicator	—	—	—
Oil level indicator	—	—	—
Sound pressure level	—	—	—
Total transport weight	—	—	—
Total oil volume	—	—	—
Paint system	—	—	—

Notes

The main MV transformer 3110-XFR-01 is an oil-immersed power transformer with a voltage ratio of 23/34.5 kV. It is rated 30 MVA under ONAN cooling and 45 MVA under ONAF cooling, with an impedance of 8.0%. An on-load tap changer provides a regulation range of ±10% in 17 positions at 1.25% per step. The transformer is designed, manufactured and tested in accordance with ANSI C57.12.00. Its connection to the plant electrical system is shown on single line diagram QAZ-5200-265-E-001_B.

17. 3110-XFR-02 AUXILIARY STATION TRANSFORMER

Auxiliary transformer supplying the 6.6 kV auxiliary bus.

Parameter	Symbol	Value	Basis
Transformer tag	Tag	3110-XFR-02	Equipment list
Service	Type	Auxiliary station transformer	Equipment list
Rated power	S	6.0 MVA	Transformer rating
Voltage ratio	V_1 / V_2	34.5 kV / 6.6 kV	Winding voltages
Impedance	Z	7.0%	Impedance at rated power
Tap range	Taps	±2 x 2.5% (5 positions)	Tap changer
Design standard	Std	ANSI C57.12.00	Standards list
Cooling class	—	—	—
Insulation fluid type	—	—	—
Oil volume	—	—	—
Oil preservation system	—	—	—
Winding material	—	—	—
HV winding connection	—	—	—
LV winding connection	—	—	—
Vector group	—	—	—
HV bushing type	—	—	—
LV bushing type	—	—	—
Neutral bushing	—	—	—
Basic insulation level	—	—	—
Insulation class	—	—	—
No-load losses	—	—	—
Load losses	—	—	—
Temperature rise	—	—	—
Hot spot limit	—	—	—
Tap changer type	—	—	—
Tap changer operation	—	—	—
Core construction	—	—	—
Tank construction	—	—	—
Conservator type	—	—	—
Pressure relief device	—	—	—
Buchholz relay	—	—	—
Winding temperature indicator	—	—	—
Oil temperature indicator	—	—	—
Oil level gauge	—	—	—
Sound level	—	—	—
Total mass	—	—	—
Shipping mass	—	—	—
Grounding method	—	—	—

Notes

The auxiliary station transformer 3110-XFR-02 is rated 6.0 MVA and steps the incoming supply down from 34.5 kV to 6.6 kV. Its impedance is 7.0%, which limits fault current on the downstream switchgear. The transformer has a tap changer with a range of ±2 x 2.5% in five positions. Design, manufacture and testing comply with ANSI C57.12.00. The secondary winding feeds the 6.6 kV auxiliary bus, which supplies motor 3220-MTR-01 and other auxiliary plant loads.

18. 3220-MTR-01 DRIVE COOLING-WATER PUMP MOTOR

Electrical data for the drive cooling-water pump motor in the 3220 auxiliary area.

Parameter	Symbol	Value	Basis
Motor tag	Tag	3220-MTR-01	Equipment list
Service	Type	Drive cooling-water pump motor	Equipment list
Rated power	P	1,250 kW	Motor rating
Supply voltage	V	6.6 kV	Auxiliary bus voltage
Power factor	pf	0.88	Design basis
Efficiency	η	0.96	Design basis
Full-load current	I_FL	129.4 A	I = P / (√3 × V × pf × η)
Motor standard	Std	IEC 60034-1	Standards list
Motor standard, North American	Std	NEMA MG 1	Standards list
Cooling method	—	—	—
Insulation class	—	—	—
Temperature rise class	—	—	—
Enclosure protection degree	—	—	—
Drive end bearing	—	—	—
Non-drive end bearing	—	—	—
Bearing lubrication method	—	—	—
Winding connection	—	—	—
Starting method	—	—	—
Locked rotor current	—	—	—
Locked rotor torque	—	—	—
Breakdown torque	—	—	—
Vibration limit	—	—	—
Noise level	—	—	—
Space heater rating	—	—	—
Winding temperature detectors	—	—	—
Bearing temperature detectors	—	—	—
Frame size	—	—	—
Mounting arrangement	—	—	—
Rotation direction	—	—	—
Duty type	—	—	—
Service factor	—	—	—
Synchronous speed	—	—	—
Rated speed	—	—	—
Number of poles	—	—	—
Shaft coupling type	—	—	—
Terminal box location	—	—	—
Motor mass	—	—	—
Ambient temperature range	—	—	—
Altitude rating	—	—	—

Notes

The drive cooling-water pump motor 3220-MTR-01 is rated 1,250 kW at 6.6 kV, with a power factor of 0.88 and an efficiency of 0.96 at rated load. The full-load current is 129.4 A. The motor is designed, manufactured and tested in accordance with IEC 60034-1 and NEMA MG 1. Auxiliary transformer 3220-XFR-01, rated 0.8 MVA, 6.6 kV / 380 V, with an impedance of 5.0%, forms part of the 3220 auxiliary area.

18.1 3220-XFR-01 AUXILIARY TRANSFORMER

Tag 3220-XFR-01. Auxiliary transformer, 0.8 MVA, voltage ratio 6.6 kV / 380 V, impedance 5.0%.

19. 3220-MCC-01 MOTOR CONTROL CENTER

Load and supply data for the motor control center in the 3220 auxiliary area.

Parameter	Symbol	Value	Basis
MCC tag	Tag	3220-MCC-01	Equipment list
Service	Type	Motor control center	Equipment list
Load power	P	450 kW	Load rating
Supply voltage	V	380 V	LV bus voltage
Power factor	pf	0.85	Design basis
Efficiency	η	0.94	Design basis
Full-load current	I_FL	855.7 A	I = P / (√3 × V × pf × η)
Enclosure type	—	—	—
Enclosure protection degree	—	—	—
Form of separation	—	—	—
Main busbar material	—	—	—
Busbar rated current	—	—	—
Busbar short-circuit rating	—	—	—
Neutral busbar size	—	—	—
Earth busbar material	—	—	—
Incomer breaker type	—	—	—
Incomer trip unit	—	—	—
Feeder breaker type	—	—	—
Motor starter type	—	—	—
Contactor utilization category	—	—	—
Overload relay type	—	—	—
Control voltage source	—	—	—
Metering instrument type	—	—	—
Current transformer ratio	—	—	—
Voltage transformer ratio	—	—	—
Protection relay type	—	—	—
Earthing system type	—	—	—
Cable entry location	—	—	—
Gland plate material	—	—	—
Paint finish color	—	—	—
Ambient design temperature	—	—	—
Space heater supply	—	—	—
Internal arc classification	—	—	—
Withdrawable unit type	—	—	—
Communication protocol	—	—	—
Interlocking scheme	—	—	—
Spare feeder provision	—	—	—
Mounting arrangement	—	—	—
Nameplate material	—	—	—

Notes

The motor control center 3220-MCC-01 supplies a connected load of 450 kW at a nominal voltage of 380 V. The design basis uses a power factor of 0.85 and an efficiency of 0.94, which gives a full-load current of 855.7 A for sizing the incoming feeder and main busbar. The mill local control panel 3210-LCP-01 is located in the mill drive area and is shown for interface only, because no electrical rating is given for it in this issue.

19.1 3210-LCP-01 MILL LOCAL CONTROL PANEL

Tag 3210-LCP-01, mill local control panel, 3210 drive area. No electrical values for this panel are given in this issue.

20. CALCULATION SUMMARY: FULL-LOAD CURRENTS

Full-load current I = P / (√3 × V × pf × η). The values below agree with the workbook summary on the Index sheet, cells H31 to H34.

Tag	Bus	Voltage	Rated power (kW)	Full-load current (A)
3210-MTR-01	34.5 kV drive bus	34.5 kV	16,500	299.6
3210-MTR-02	34.5 kV drive bus	34.5 kV	12,000	217.9
3220-MTR-01	6.6 kV auxiliary bus	6.6 kV	1,250	129.4
3220-MCC-01	380 V LV bus	380 V	450	855.7
Load tag	—	—	—	—
Load description	—	—	—	—
Supply voltage	—	—	—	—
Rated output power	—	—	—	—
Power factor	—	—	—	—
Motor efficiency	—	—	—	—
Full-load current	—	—	—	—
Calculation formula	—	—	—	—
Phase count	—	—	—	—
System frequency	—	—	—	—
Apparent power	—	—	—	—
Reactive power	—	—	—	—
Active power input	—	—	—	—
Voltage tolerance	—	—	—	—
Frequency tolerance	—	—	—	—
Service factor	—	—	—	—
Duty type	—	—	—	—
Starting method	—	—	—	—
Starting current ratio	—	—	—	—
Load factor	—	—	—	—
Diversity factor	—	—	—	—
Ambient temperature	—	—	—	—
Altitude derating factor	—	—	—	—
Cable sizing current	—	—	—	—
Protection setting basis	—	—	—	—
Supply bus reference	—	—	—	—
Feeder reference	—	—	—	—
Calculation method	—	—	—	—
Data source	—	—	—	—
Workbook reference cell	—	—	—	—
Result rounding	—	—	—	—
Verification status	—	—	—	—
Checked by	—	—	—	—

Notes

The full-load current calculation was carried out for the four loads supplied from the gearless mill drive distribution system. Motor 3210-MTR-01 at 34.5 kV has a full-load current of 299.6 A, and motor 3210-MTR-02 at 34.5 kV has a full-load current of 217.9 A. Motor 3220-MTR-01 at 6.6 kV draws 129.4 A, and motor control centre 3220-MCC-01 at 380 V draws 855.7 A. These values agree with cells H31 to H34 on the Index sheet of the companion workbook.